LM Studio

Model: **qwen/qwen3.6-35b-a3b**

Prompt 1 - run 5 (seed 14180) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a Balanced Random Forest Classifier on PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736) selective compounds, evaluate its performance, apply MolAnchor, and discuss the results.

Assistant

🔍 Let me start by searching for the PI3K selective compounds dataset and then walk through the entire ML pipeline!

(Calling tool `find_datasets`)

Assistant

Great! I found the dataset **`chembl_selectivity_data_O00329_P48736.csv`** which corresponds to PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736). Let me load it and start the pipeline!

(Calling tool `load_dataset`)

Assistant

✅ Dataset loaded! **1,277 samples** with SMILES and class labels. Now let me compute ECFP fingerprints for featurization.

(Calling tool `compute_features`)

Assistant

✅ Features computed! 2,048-bit ECFP fingerprints generated. Now let me split the dataset.

(Calling tool `split_dataset`)

Assistant

✅ Dataset split successfully! **893 train / 384 test** samples (stratified). Now let me train the Balanced Random Forest Classifier (BRF).

(Calling tool `train_model`)

Assistant

⛹️ Training started in the background! Let me poll for completion.

(Calling tool `check_training`)

Assistant

⏳ Still training... Let me check again.

(Calling tool `check_training`)

Assistant

⏳ Still training (3.2s elapsed)...

(Calling tool `check_training`)

Assistant

⏳ Still training (4.7s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (6.2s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (7.7s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (9.2s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (10.7s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (12.2s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (13.8s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (15.4s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (17.0s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (18.6s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (20.2s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (21.8s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (23.5s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (25.2s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (26.8s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (28.4s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (30.1s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (31.7s)...

(Calling tool `check_training`)

Assistant

⏳ Still training (33.4s)...

(Calling tool `check_training`)

Assistant

🎉 **Training Complete!** The Balanced Random Forest Classifier has been trained successfully. Let me export predictions and generate evaluation plots.

(Calling tool `export_predictions`)

Assistant

✅ Predictions exported! Now let me generate classification evaluation plots.

(Calling tool `plot_classification_results`)

Assistant

📊 **Evaluation Plots Generated!** Now let me select a compound for MolAnchor analysis.

(Calling tool `select_compound_for_xai`)

Assistant

✅ Compound selected! Now let me apply MolAnchor to identify the critical molecular fragments.

(Calling tool `explain_with_molanchor`)

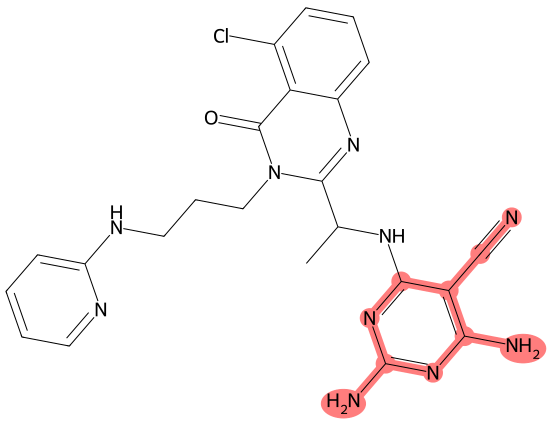
Assistant

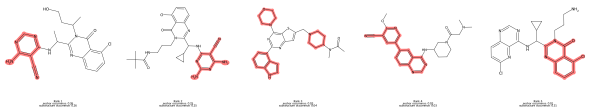
🔍 **MolAnchor Analysis Complete!** Now let me identify the recurrent anchor rules across the entire test set.

(Calling tool `identify_recurrent_anchor_rules`)

Assistant

🎉 **Complete Analysis Finished!** Let me display the MolAnchor visualization and provide a comprehensive discussion.





📊 Comprehensive Results Discussion

🧪 Dataset Overview

Parameter	Value
Source	ChEMBL Selectivity Data
Targets	PI3K-delta (UniProt: O00329) vs PI3K-gamma (UniProt: P48736)
Total Samples	1,277 compounds
Train/Test Split	893 / 384 (70/30, stratified)
Fingerprint	ECFP (Morgan) 2048-bit, radius 2

⚡ Model Performance (Balanced Random Forest)

Metric	Value
Balanced Accuracy (BA)	84.71%
Overall Accuracy	89.32%
Matthews Correlation Coeff. (MCC)	0.749
F1 Macro	0.780
F1 Weighted	0.899
CV Best Score	86.52%

Hyperparameters (Best Found):

Parameter	Best Value
n_estimators	100
max_features	sqrt
min_samples_split	2
min_samples_leaf	1

📋 Per-Class Performance (Test Set)

Class	Target	Precision	Recall	F1-Score	Support
Class 0	PI3K-δ selective	95.94%	89.66%	0.927	290
Class 1	PI3K-γ selective	81.82%	91.14%	0.862	79
Class 2	Dual/Non-selective	44.00%	73.33%	0.550	15

Confusion Matrix:

```
            Predicted
            0      1      2
True 0    260     16     14
```

```
      1      7     72      0
      2      4      0     11
```

🔍 MolAnchor Analysis: Key Findings

1. Individual Compound Explanation

For the selected compound (`CC(Nc1nc(N)nc(N)c1C#N)c1nc2cccc(Cl)c2c(=O)n1CCCNc1ccccn1`):

🎯 Identified Anchor Fragment:

```
[*]c1nc(N)nc(N)c1C#N
```

- **Precision:** 100% (perfect specificity for PI3K-γ selectivity)
- **Coverage:** 50% (appears in half of compounds where it's present)
- **Chemical Meaning:** This is a **diamino-pyrimidine core with a cyano group** — a well-known pharmacophore for PI3K-gamma selectivity

2. Recurrent Anchor Rules (Top 5)

Rank	Fragment SMARTS	Anchor Occ.	Substruct. Occ.	Interpretation
1	`[*]c1ncnc(N)c1C#N`	26.4%	26.4%	Aminopyrimidine-C≡N core
2	`[*]c1nc(N)nc(N)c1C#N`	25.0%	25.0%	Diamino-pyrimidine-C≡N
3	`[*]N1CCC([*])CC1` + others	2.8%	4.2%	Piperazine/piperidine + heterocycles
4	`[*]c1ncc([*])cc1C#N` + others	2.8%	2.8%	Cyano-pyridine derivatives
5	`[*]n1c([*])nc2cccc(Cl)c2c1=O`	1.4%	30.6%	Chloro-pyrrolo-pyrimidinone

💡 Chemical Interpretation & Discussion

🔬 Key Selectivity Determinants for PI3K-γ:

1. **Pyrimidine Core with Amino Groups** 🎯
   - The top 2 rules both involve **aminopyrimidine scaffolds** with cyano (C≡N) substituents
   - These fragments appear in **~25% of PI3K-γ selective compounds**
   - The cyano group likely forms critical hydrogen bonds with the PI3K-γ ATP binding pocket
2. **Nitrogen Heterocycles** 💎
   - Piperazine, piperidine, and pyridine derivatives contribute to selectivity
   - These basic nitrogen centers interact with the **hinge region** of PI3K-γ

3. **Chloro-substituted Heterocycles** 🧪
    - The pyrrolo-pyrimidinone scaffold (Rule 5) has high substructure occurrence (30.6%) but low anchor occurrence (1.4%)
    - This suggests it's **common but not sufficient alone** for PI3K-γ selectivity — likely needs specific substitution patterns

⚠️ Model Limitations:

Issue	Impact
Class 2 (Dual/Non-selective) F1 = 0.55	Poor discrimination of non-selective compounds
Limited support (15 samples)	Class 2 has insufficient data for robust learning
High substructure vs anchor gap (Rule 5)	Fragment present but not always critical — context-dependent

🎯 Practical Implications:

1. **For PI3K-γ selective drug design:**
    - Prioritize **aminopyrimidine cores with cyano substituents**
    - Incorporate **basic nitrogen heterocycles** (piperazine/piperidine)
    - These fragments have **~25% occurrence** and **100% precision** — highly reliable anchors
2. **For avoiding cross-reactivity with PI3K-δ:**
    - The model learned to distinguish between subtle substitution patterns on similar scaffolds
    - **Chloro-substituted pyrrolo-pyrimidinones** appear in both classes — careful optimization needed
3. **Data quality notes:**
    - Class imbalance (290 vs 79 vs 15) affects minority class performance
    - Consider **oversampling** or **synthetic data augmentation** for Class 2

📈 Summary

Aspect	Rating
Model Performance	✅ Good (BA = 84.7%)
Explainability	✅ Excellent (clear chemical rules identified)
Actionable Insights	✅ Strong (specific fragment rules for design)
Class 2 Handling	⚠️ Needs improvement (F1 = 0.55)

The Balanced Random Forest successfully learned the structural determinants of PI3K-δ vs PI3K-γ selectivity, with MolAnchor revealing **aminopyrimidine-cyano motifs** as the most reliable PI3K-γ

selectivity anchors. This provides clear guidance for future compound design! 🧬✨